Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Find Eigensolutions with Real Distinct Eigenvalues

1. The General Solution with Real Distinct Eigenvalues

Consider the homogeneous linear system of 1st-order ODEs: $\begin{aligned} x' &= -2x + y \\ y' &= x - 2y \end{aligned}$.

a. The general solution to the given system is

$$\begin{aligned} x &= -c_1 e^{-3t} + c_2 e^{-t} \\ y &= c_1 e^{-3t} + c_2 e^{-t} \end{aligned}.$$

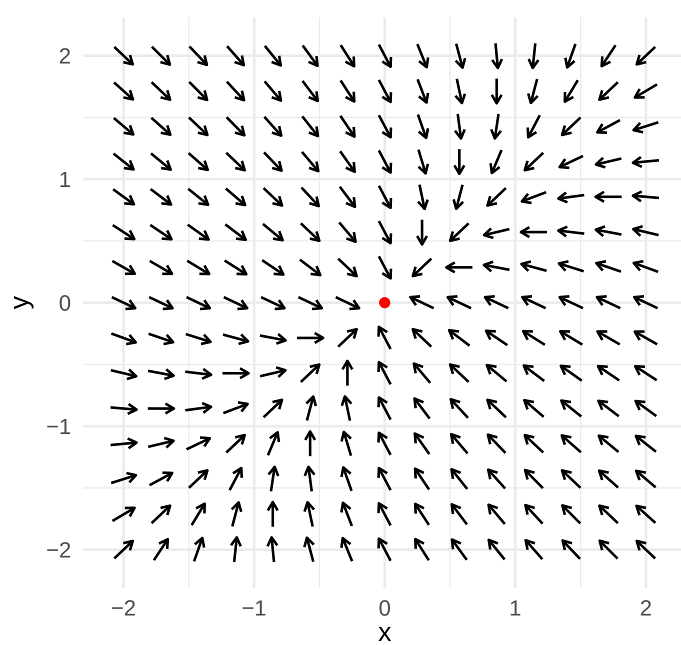
Write this solution into vector form.

b. From the vector form of the general solution in Part (a), identify the eigenvalues and eigenvectors.

c. Explain what the eigenvectors describe when viewed on a phase plane.

d. Use the properties of the eigenvalues to determine the stability of the equilibrium.

c. Draw the eigenvectors and at least four specific solution curves on the vector field Then, describe the solution behaviors around the equilibrium.



2. Finding Eigenvalues and Eigenvectors

Consider the same system of ODEs from Problem (1) in matrix-vector form: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

a. To solve the system, we need to find any eigenvalue λ with corresponding eigenvector $\vec{v} \neq \vec{0}$ so that

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \left(\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) \vec{v}$$

Explain why this is related to finding the straight-line solutions.

b. The characteristic equation can be determined using the formula

$$\det \left(\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0.$$

Use the characteristic equation to show that the eigenvalues are $\lambda_1 = -3$ and $\lambda_2 = -1$.

c. Given the first eigenvalue $\lambda_1 = -3$, we need to find its corresponding eigenvector $\vec{v}_1$. The following tasks shows you how to determine $\vec{v}_1$ directly from the coefficient matrix.

Task	Show work here
Write the equation into an augmented matrix: $\begin{array}{l} \text{row}_1 \to \\ \text{row}_2 \to \end{array} \left[\begin{array}{cc\|c} -2-\lambda_1 & 1 & 0 \\ 1 & -2-\lambda_1 & 0 \end{array}\right]$	
Row reduce the augmented matrix into the form $\begin{array}{l} \text{row}_1 \to \\ \text{row}_2 \to \end{array} \left[\begin{array}{cc\|c} a & b & 0 \\ 0 & 0 & 0 \end{array}\right]$ where a and b are values left over.	
The values in row_1 correspond to the coefficients of the line $ax + by = 0$. This is one of the straight-line solutions. Then, find a vector that is parallel to this line.	

d. Given the second eigenvalue $\lambda_2 = -1$, do the same tasks shown in Part (c) to determine the corresponding eigenvector $\vec{v}_2$.

e. Write the general solution. Then, solve for the specific solution using initial conditions $(1, 0)$.